

HIDDEN COST OF YOUR DINNER PLATE

What Foods Are Hurting the Planet the Most?



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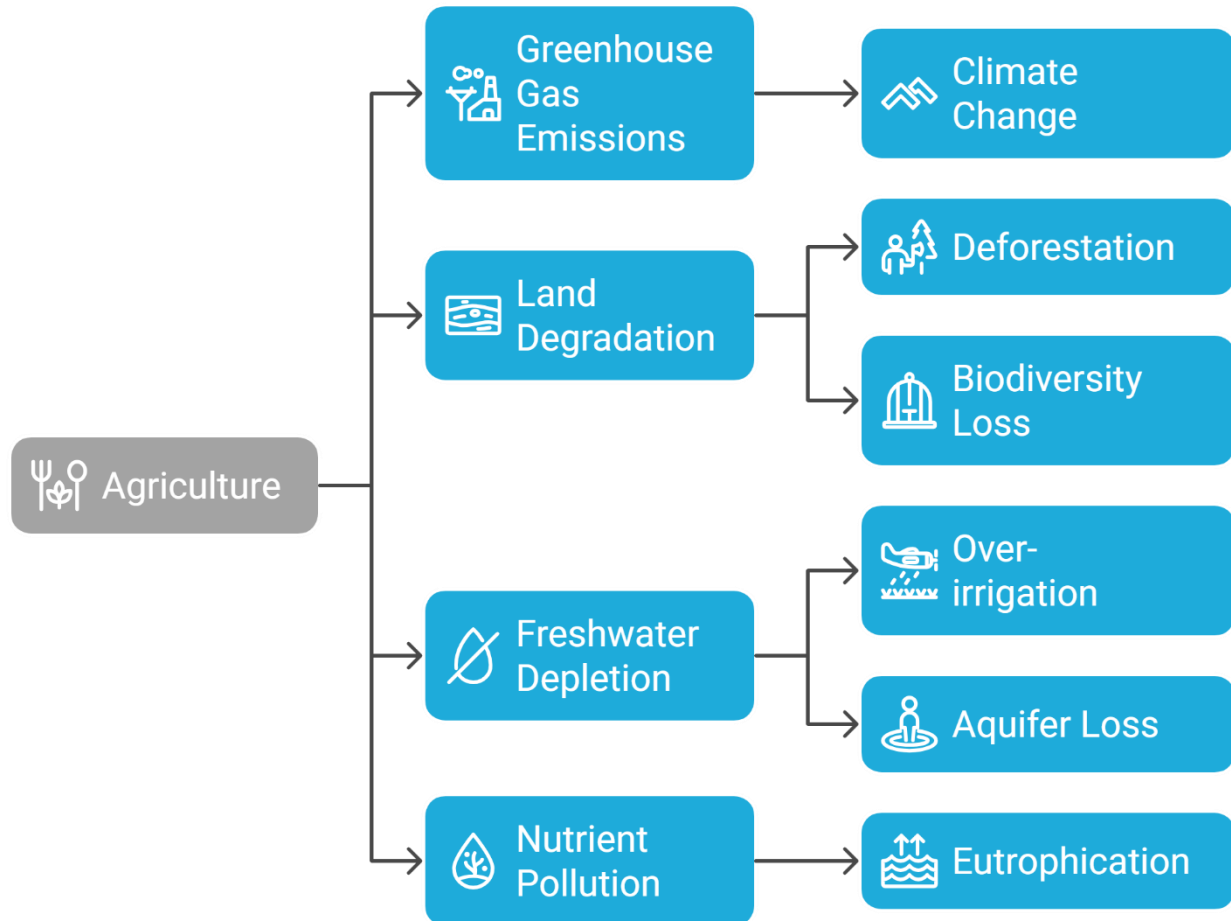
Accra, 2025



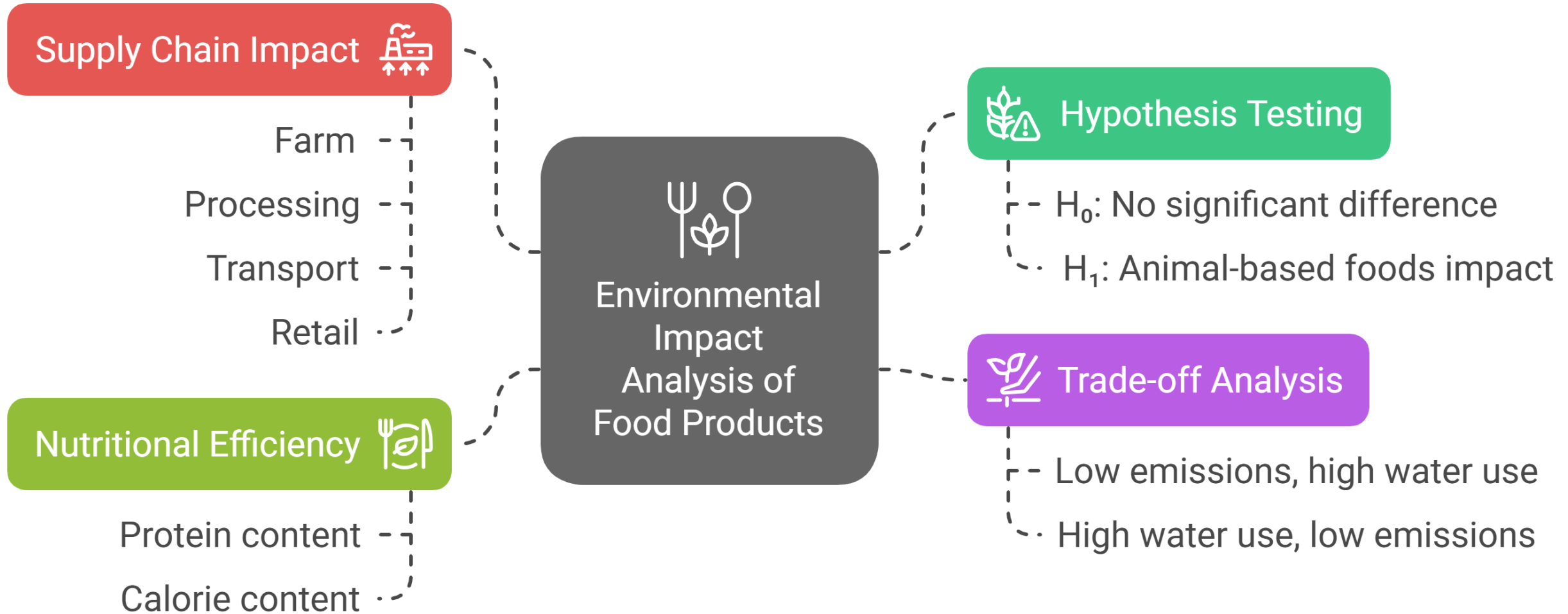
BUSINESS UNDERSTANDING

Why Food Production Matters?

- Agriculture is a major driver of environmental degradation (Adedibu, 2023; Cimini, 2021).
- Issues: **GHG emissions, land degradation, water depletion, nutrient pollution.**
- Urgent need for sustainable food systems.

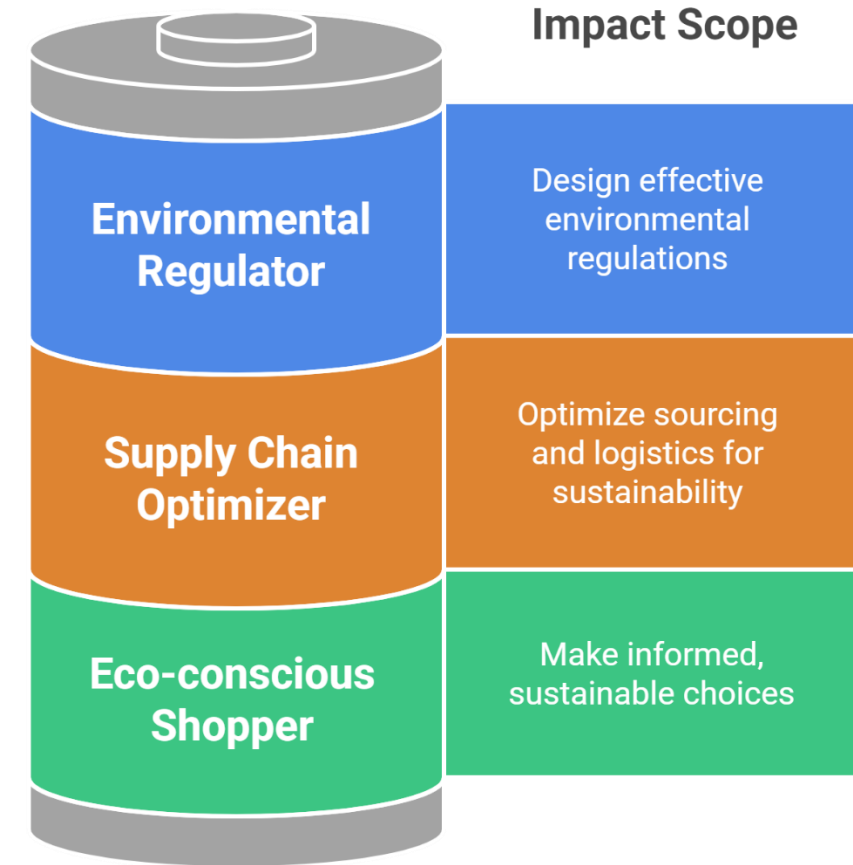


Environmental Impact of Food Production



Who Benefits From This Study?

- **Consumers:** Make informed eco-conscious food choices.
- **Businesses:** Optimize sourcing and logistics.
- **Policymakers:** Design better regulations.



Scope of Impact on Stakeholders

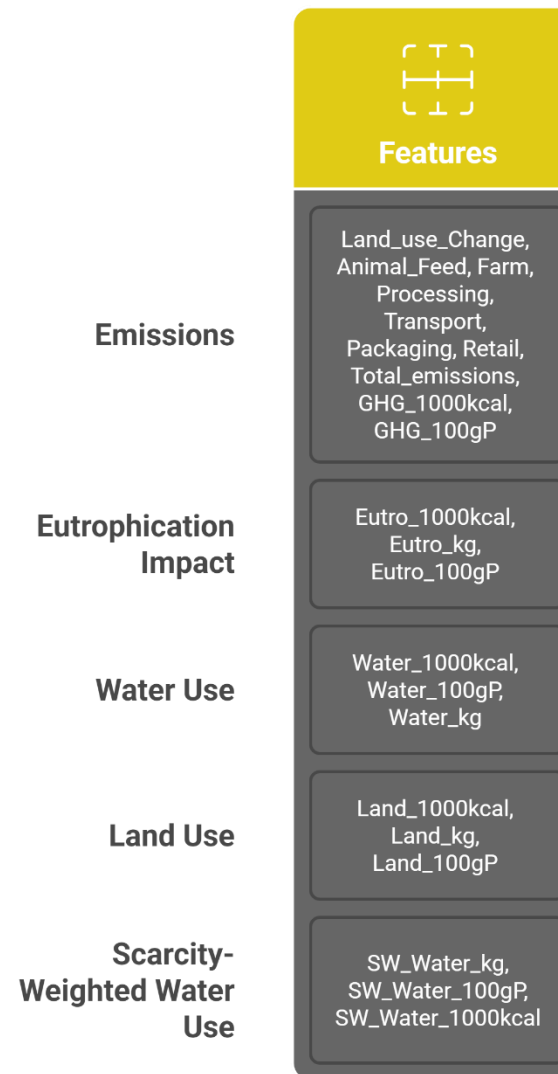
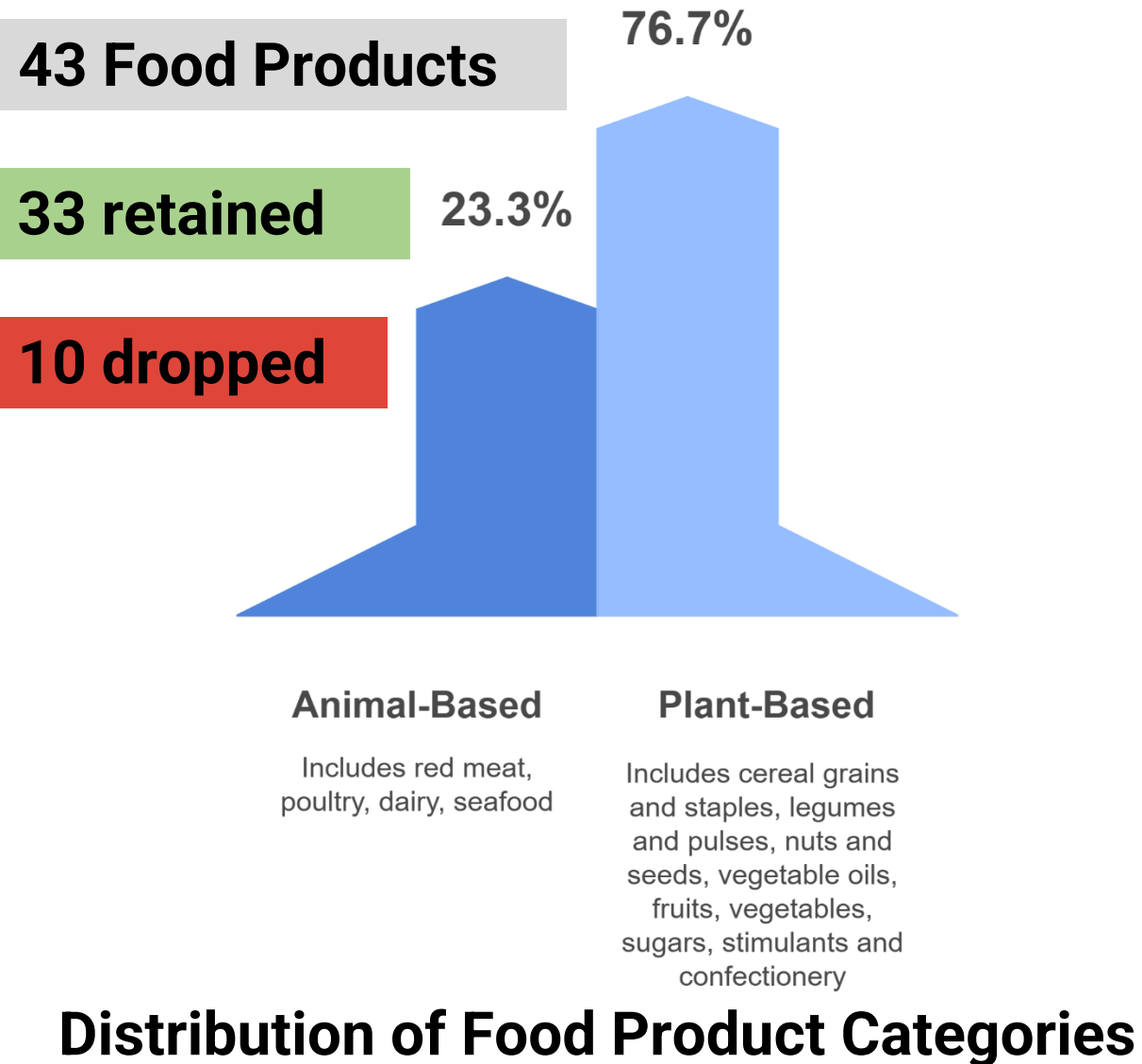
- Greenhouse Gas Emissions (**CO₂-eq**)
- Land Use (**m²**)
- Water Use (**L**)
- Eutrophication (**g PO₄-eq**)



Key Environmental Metrics

How the Metrics Inform Decisions?

Metric Format	Use Case	Example Question
Per kg	Absolute environmental impact	“Which foods emit the most CO ₂ per kg?”
Per 1000 kcal	Nutritional energy efficiency	“Which are least impactful per calorie?”
Per 100g protein	Protein-adjusted environmental cost	“Which are most efficient protein sources?”



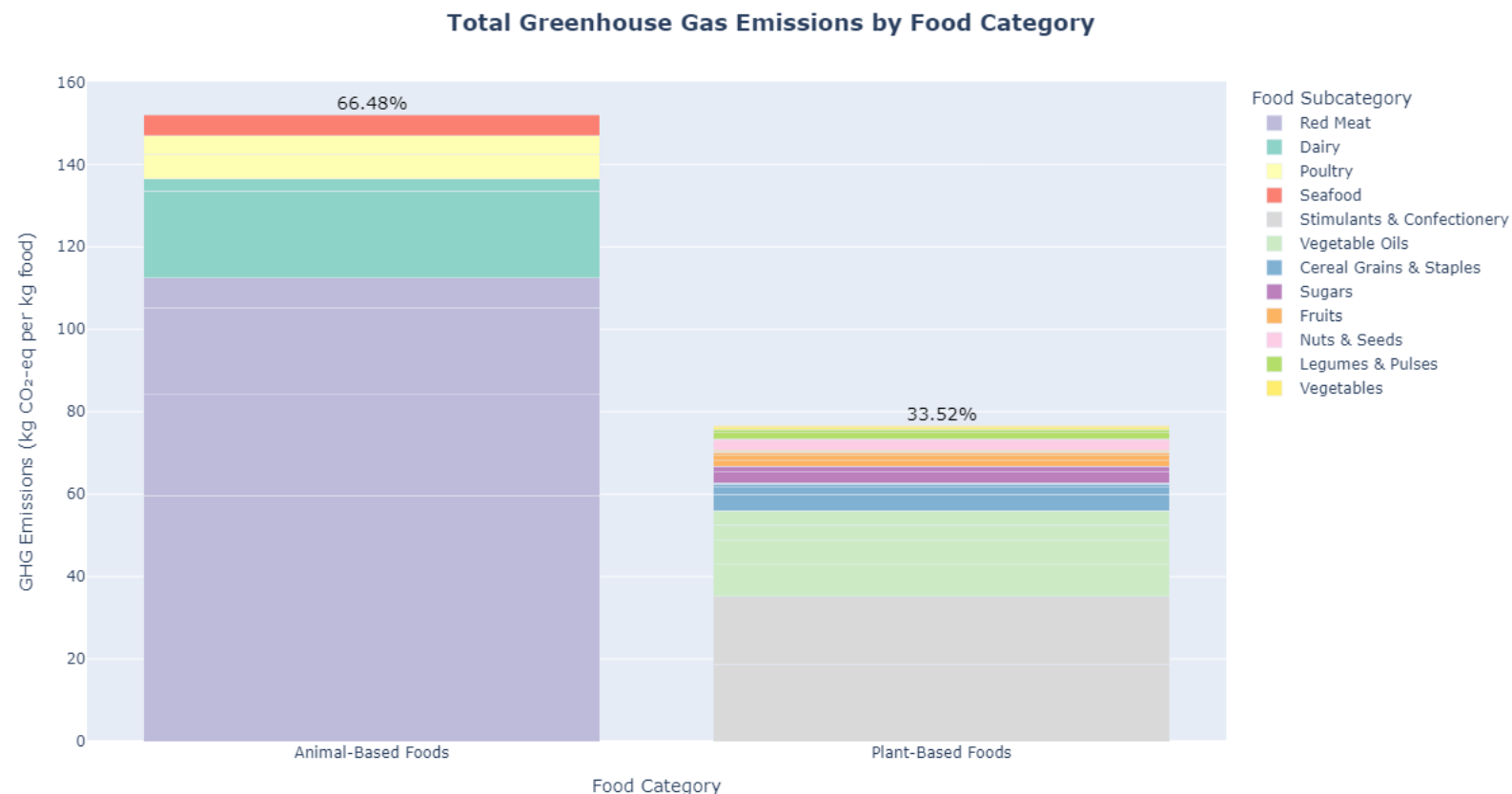
Environmental Impact Features



DATA UNDERSTANDING

Animal vs. Plant-Based: The Emissions Divide

- Determine the relative climate impact of animal-based vs. plant-based foods.

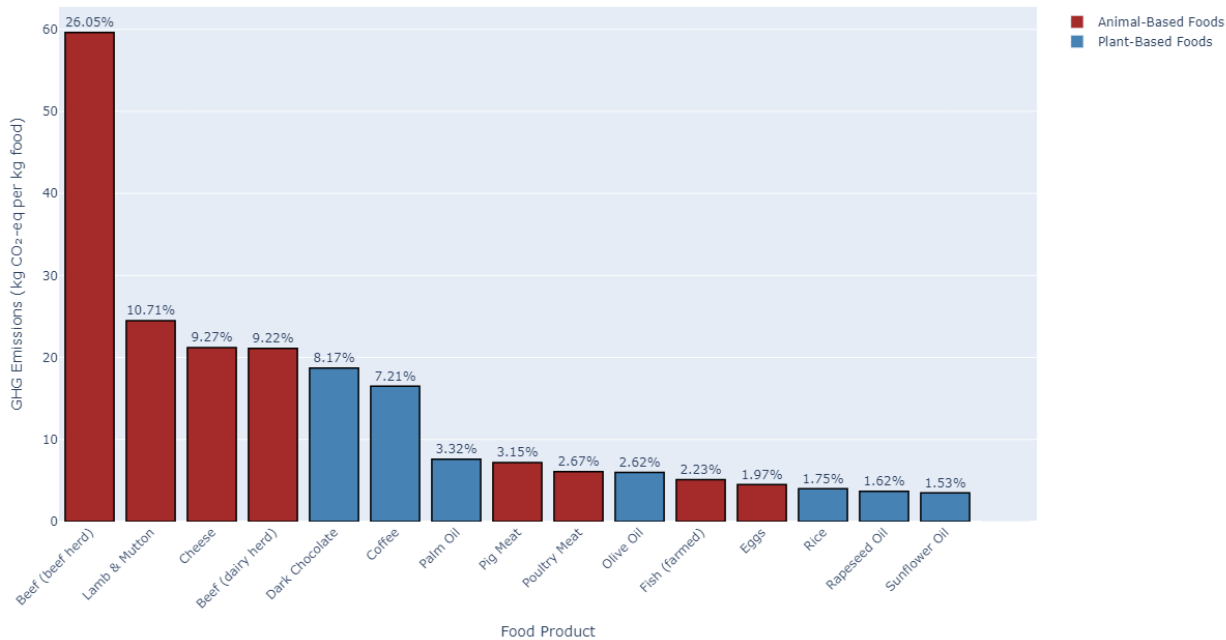


Comparison of greenhouse gas emissions (GHG) from animal-based and plant-based foods

Who Are the Worst (and Best) Climate Offenders?

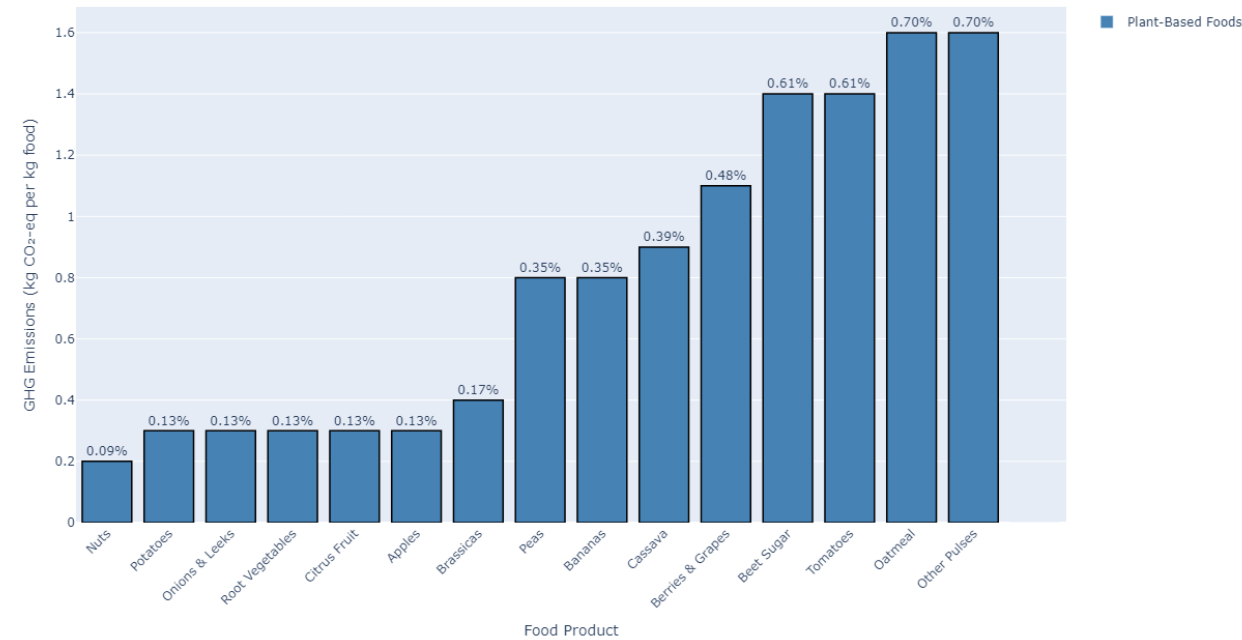
- Identify the worst offenders in terms of climate impact to promote eco-friendly diets.

Top 15 Food Products Contributing to Greenhouse Gas Emissions



Top 15 Highest GHG-Emitting Food Products (per kg food)

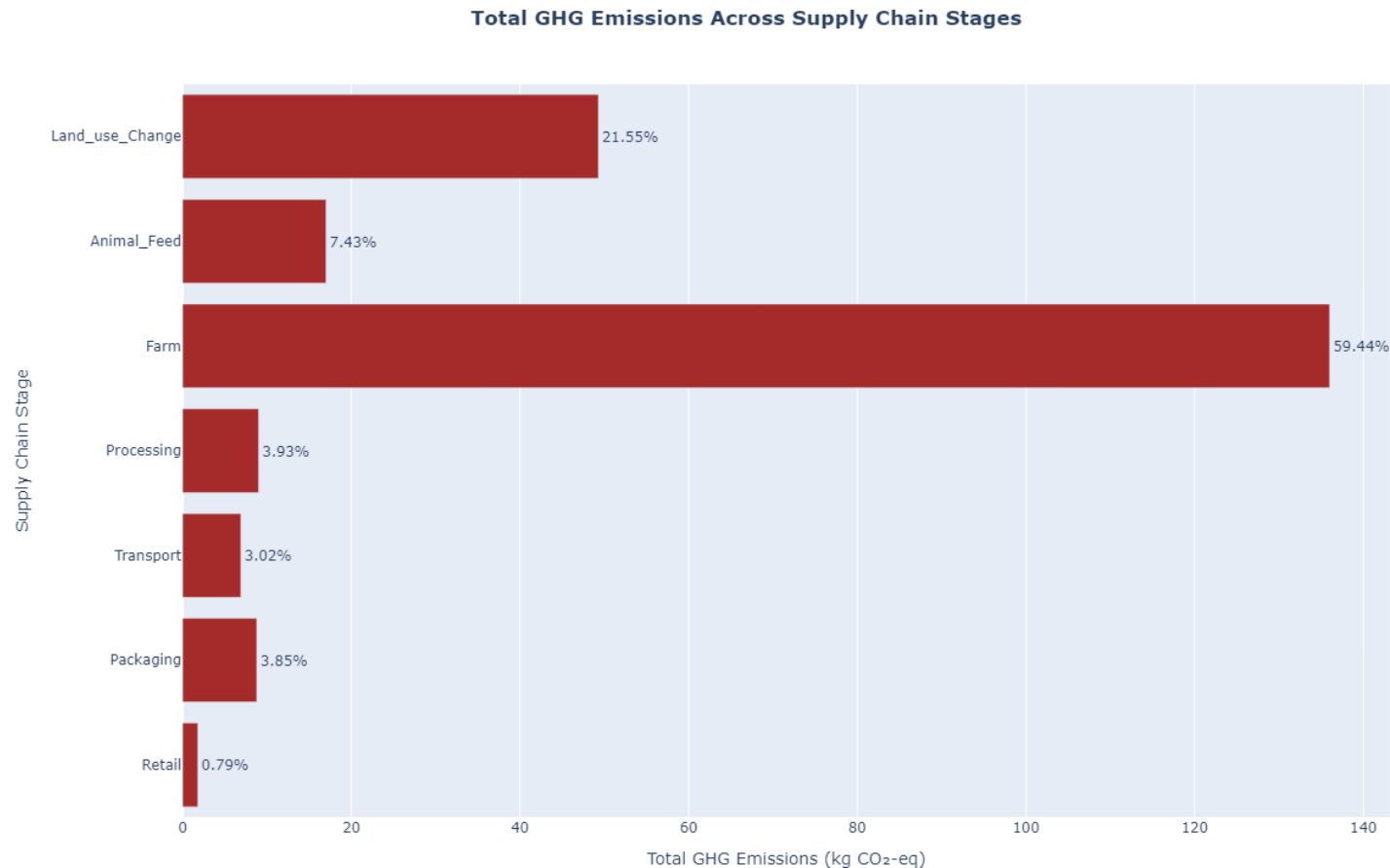
Bottom 15 Food Products Contributing to Greenhouse Gas Emissions



Bottom 15 Highest GHG-Emitting Food Products (per kg food)

Where Emissions Come From: Farm to Retail?

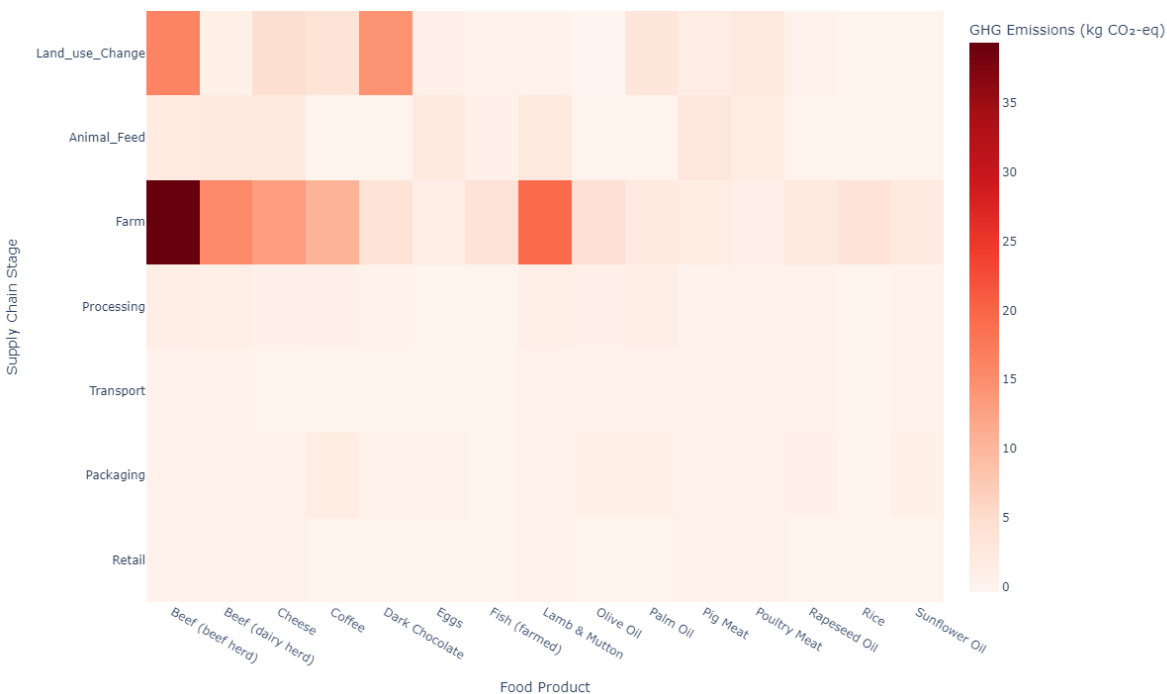
- Pinpoint which stages (e.g., farming, transport, processing) require intervention.



Proportional contribution of supply chain stages to total greenhouse gas (GHG) emissions across all food products

Where Emissions Come From: Farm to Retail?

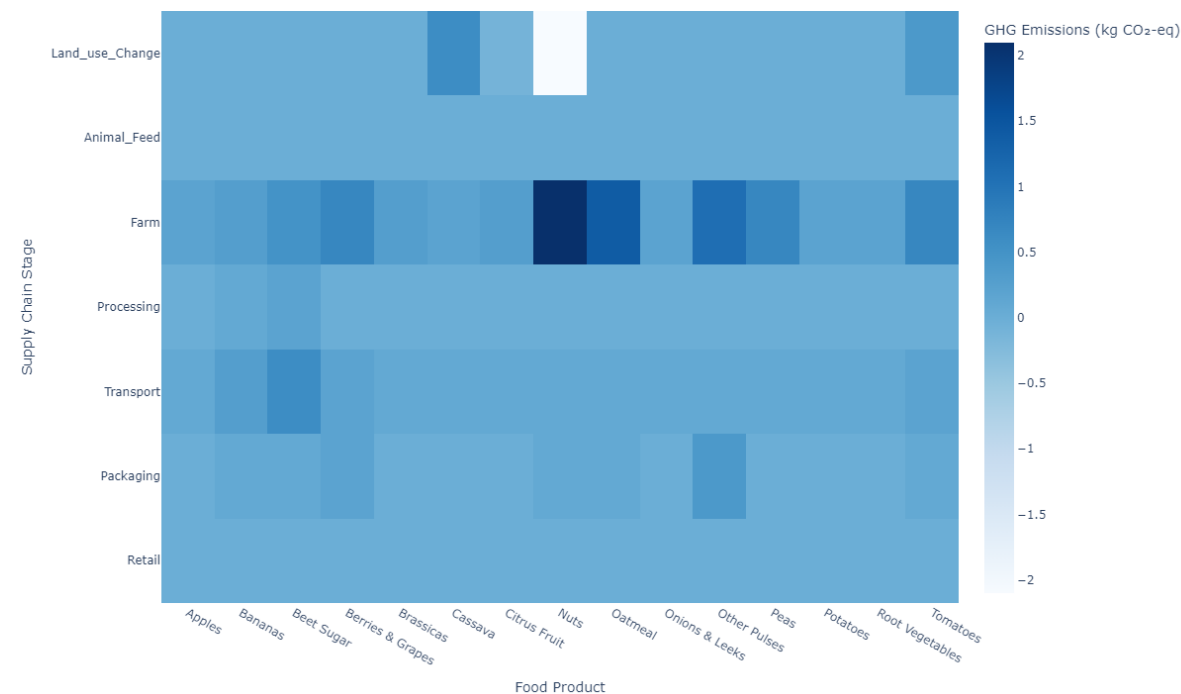
Heatmap of Supply Chain Emissions for Top 15 Contributing Food Products



GHG emissions breakdown by supply chain stage for the 15 highest-emitting food products

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Heatmap of Supply Chain Emissions for Bottom 15 Contributing Food Products



GHG emissions breakdown by supply chain stage for the 15 lowest-emitting food products

Trade-Offs Between Land Use, Water, and Emissions

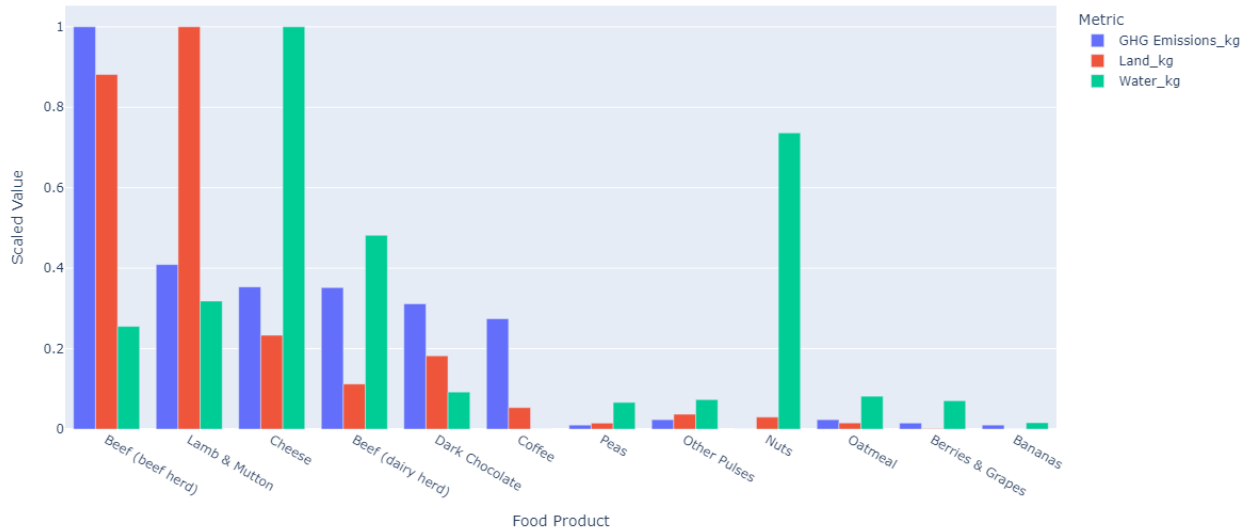
- Assess if foods with low emissions have high water use, and vice versa.



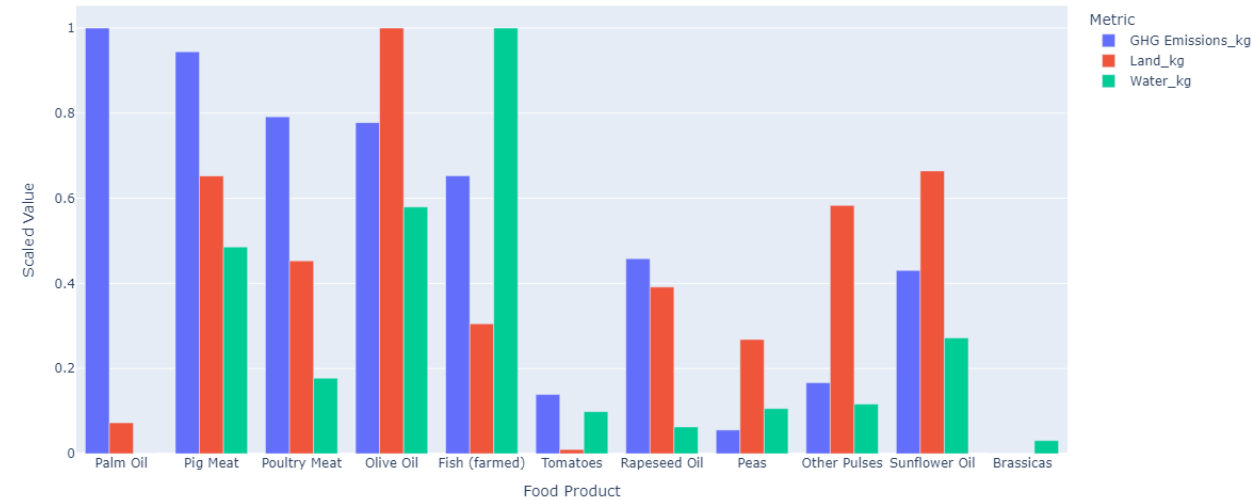
GHG emissions (y-axis) versus land use (x-axis), with bubble size representing water use, plotted per kilogram of food product

- Provide alternatives for high-impact foods to guide sustainable consumption.

A Comparison: High-Impact Foods vs. Sustainable Alternatives (Standardized Metrics)



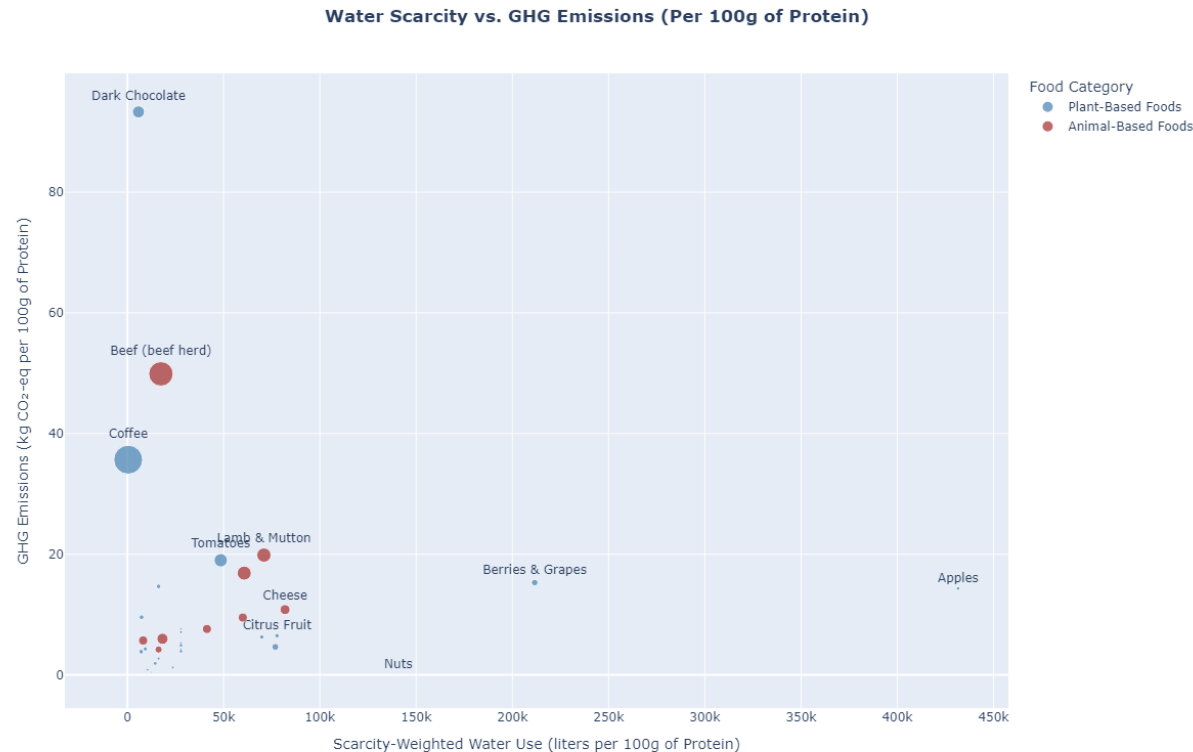
B Comparison: High-Impact Foods vs. Sustainable Alternatives (Standardized Metrics)



Comparison of high-impact foods versus more sustainable alternatives based on standardised environmental metrics

Water Scarcity and Nutritional Value

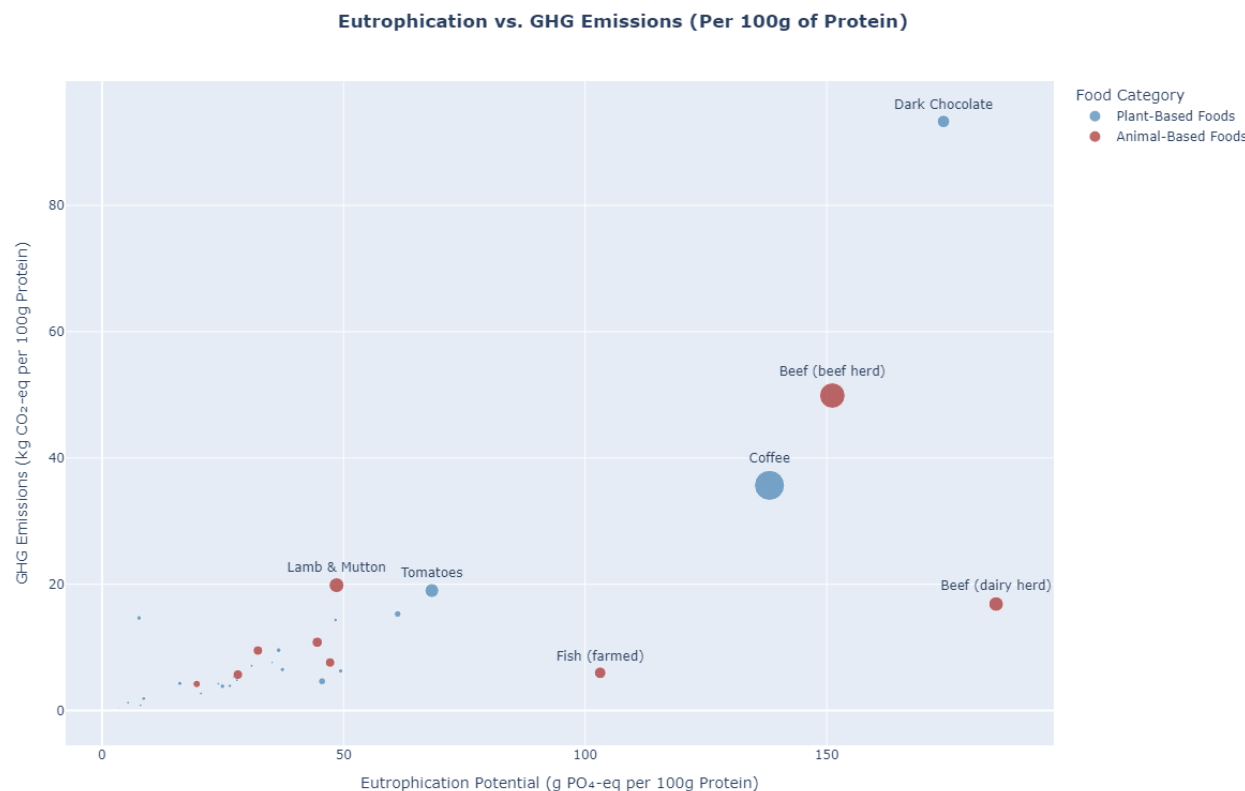
- Identify foods with high scarcity-weighted water use that may need water-efficient farming practices.



GHG emissions (y-axis) versus Scarcity-weighted water use (x-axis), plotted per 100 grams of food protein, with bubble size representing GHG emissions in 1000 kcal

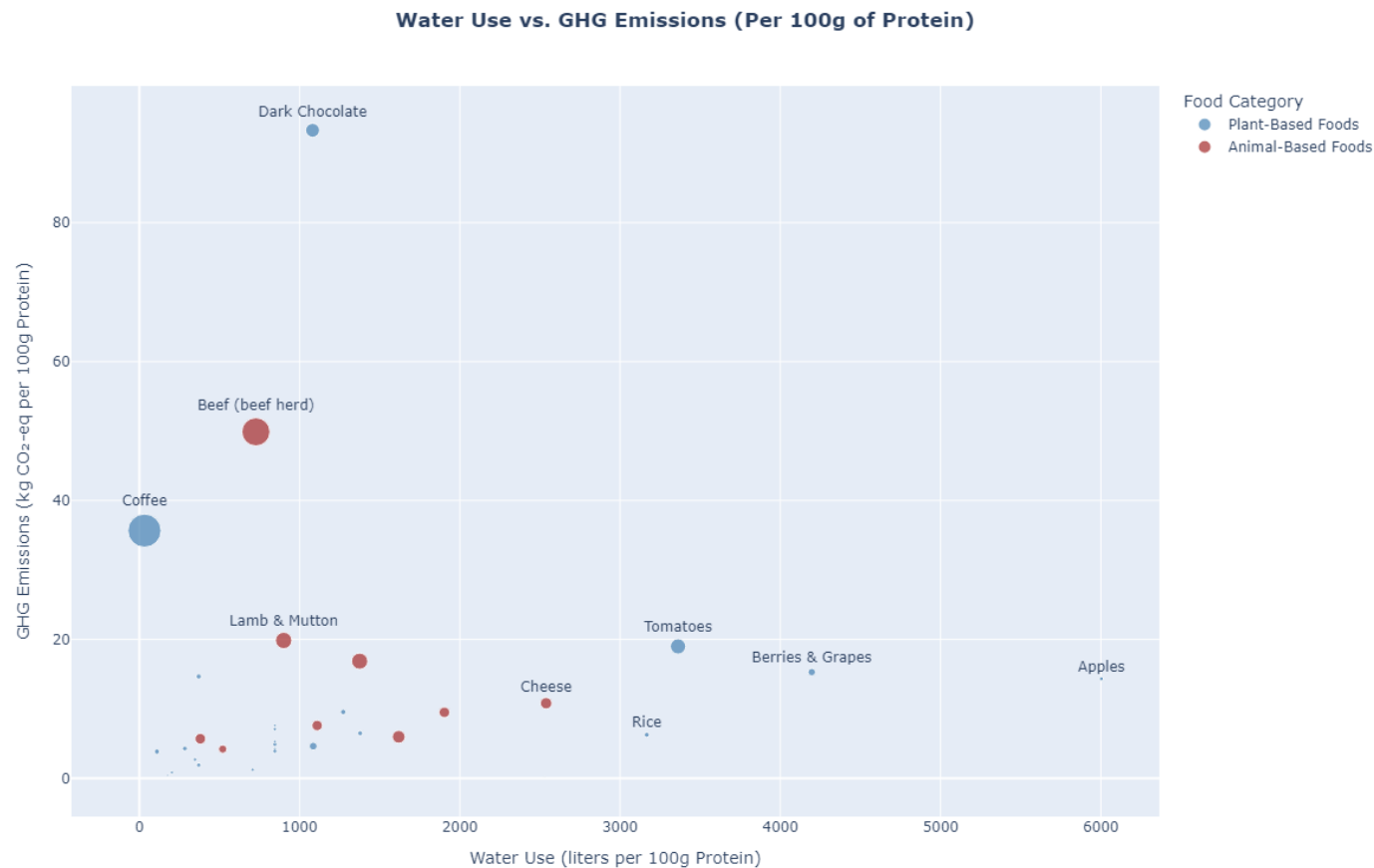
Eutrophication and Climate Burden: The Dual Threat

- Understand how nutrient runoff from agriculture (e.g., nitrogen and phosphorus use) impacts freshwater ecosystems.



Water Use vs. Climate Impact per 100g Protein

- Identify foods with excessive freshwater withdrawals and assess their sustainability.



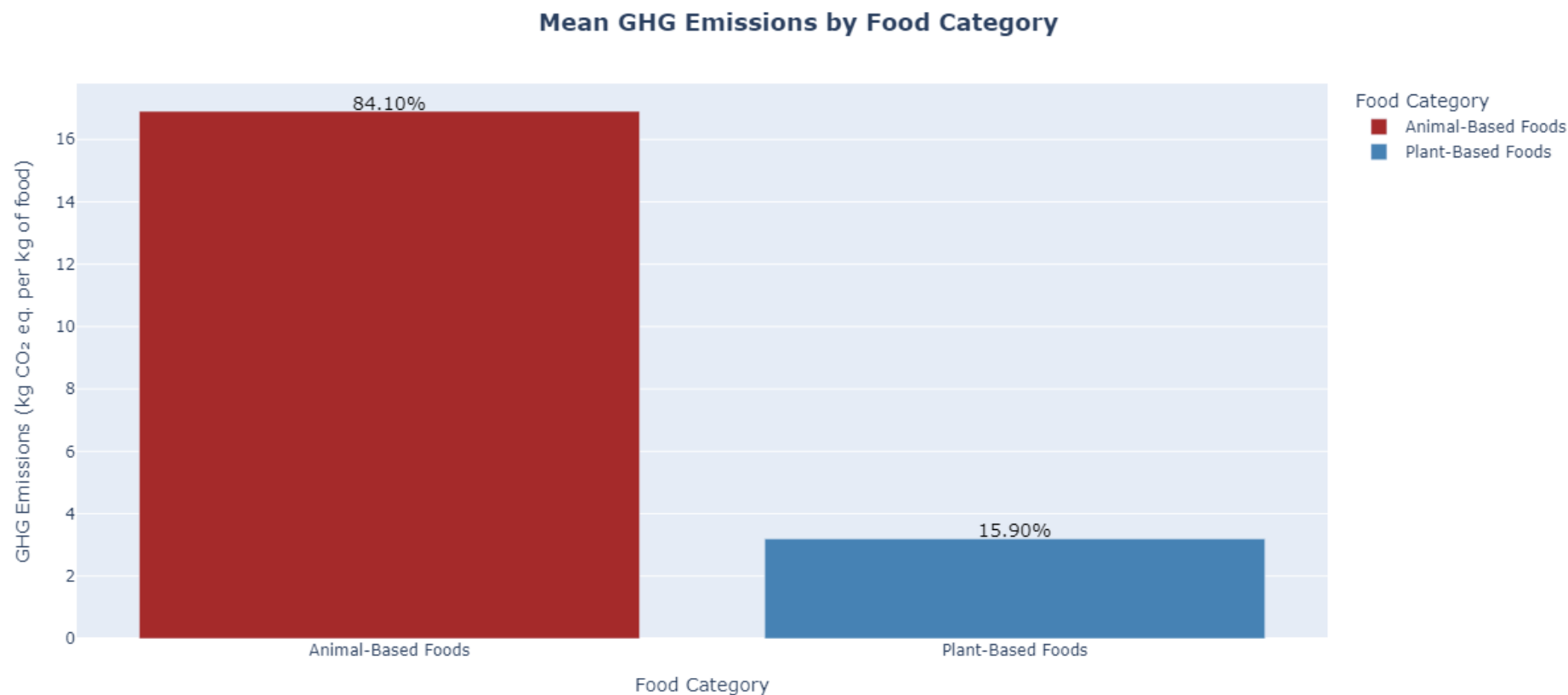
19 **GHG emissions (y-axis) versus water use (x-axis), plotted per 100 grams of food protein, with bubble size representing GHG emissions in 1000 kcal**



HYPOTHESIS TESTING

Mean emissions per kilogram of food

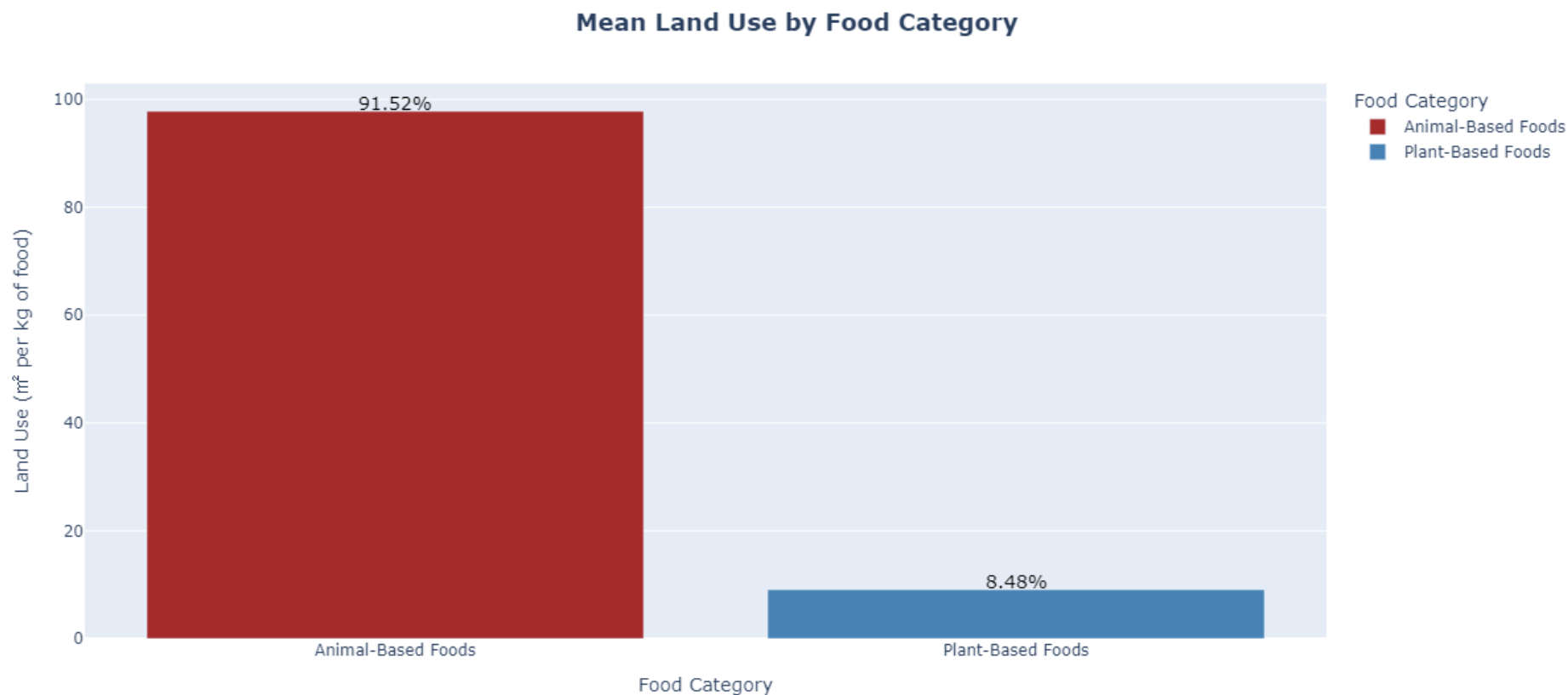
- Animal-based foods emit over 5× more greenhouse gases than plant-based foods.



Comparison of average greenhouse gas emissions (GHG) from animal-based and plant-based foods

Mean land use per kilogram of food

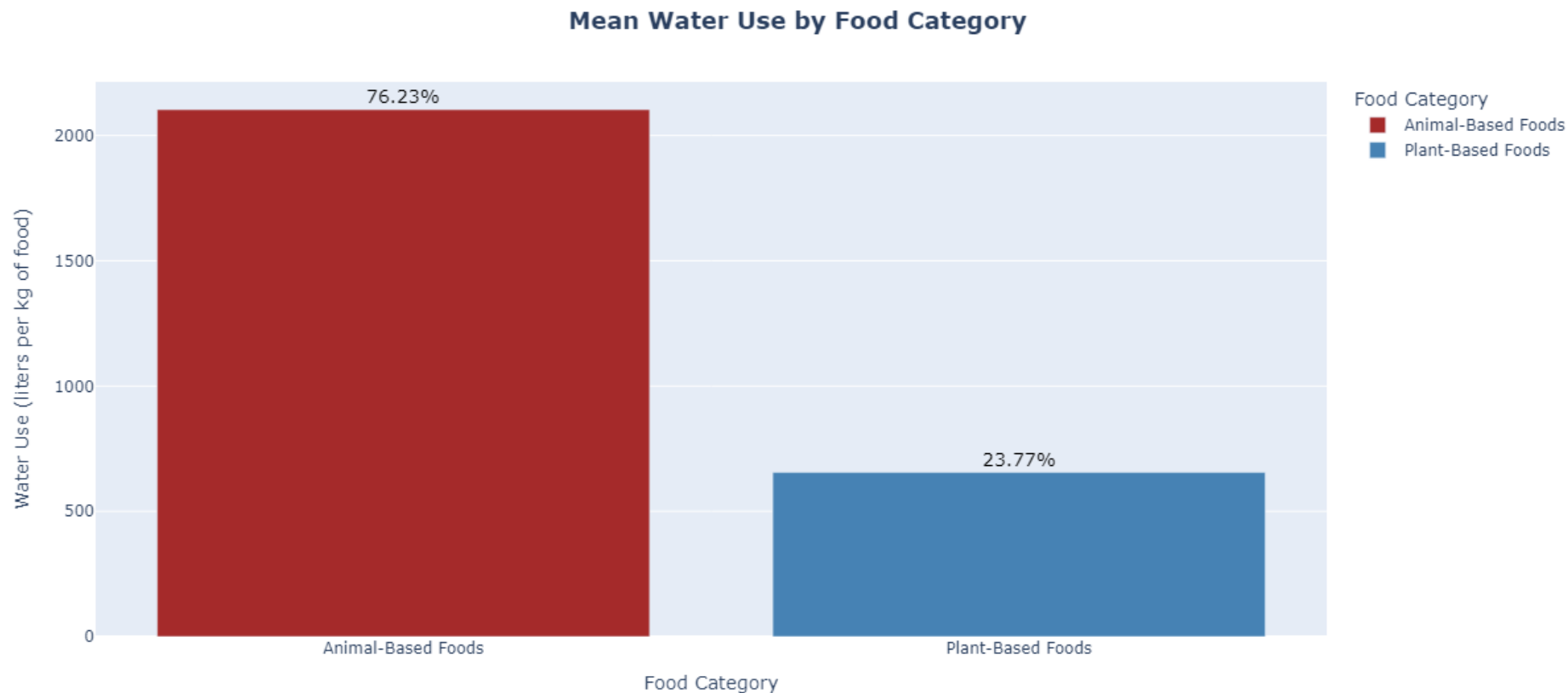
- Animal-based foods require over 10× more land than plant-based foods.



Comparison of average land use from animal-based and plant-based foods

Mean water use per kilogram of food

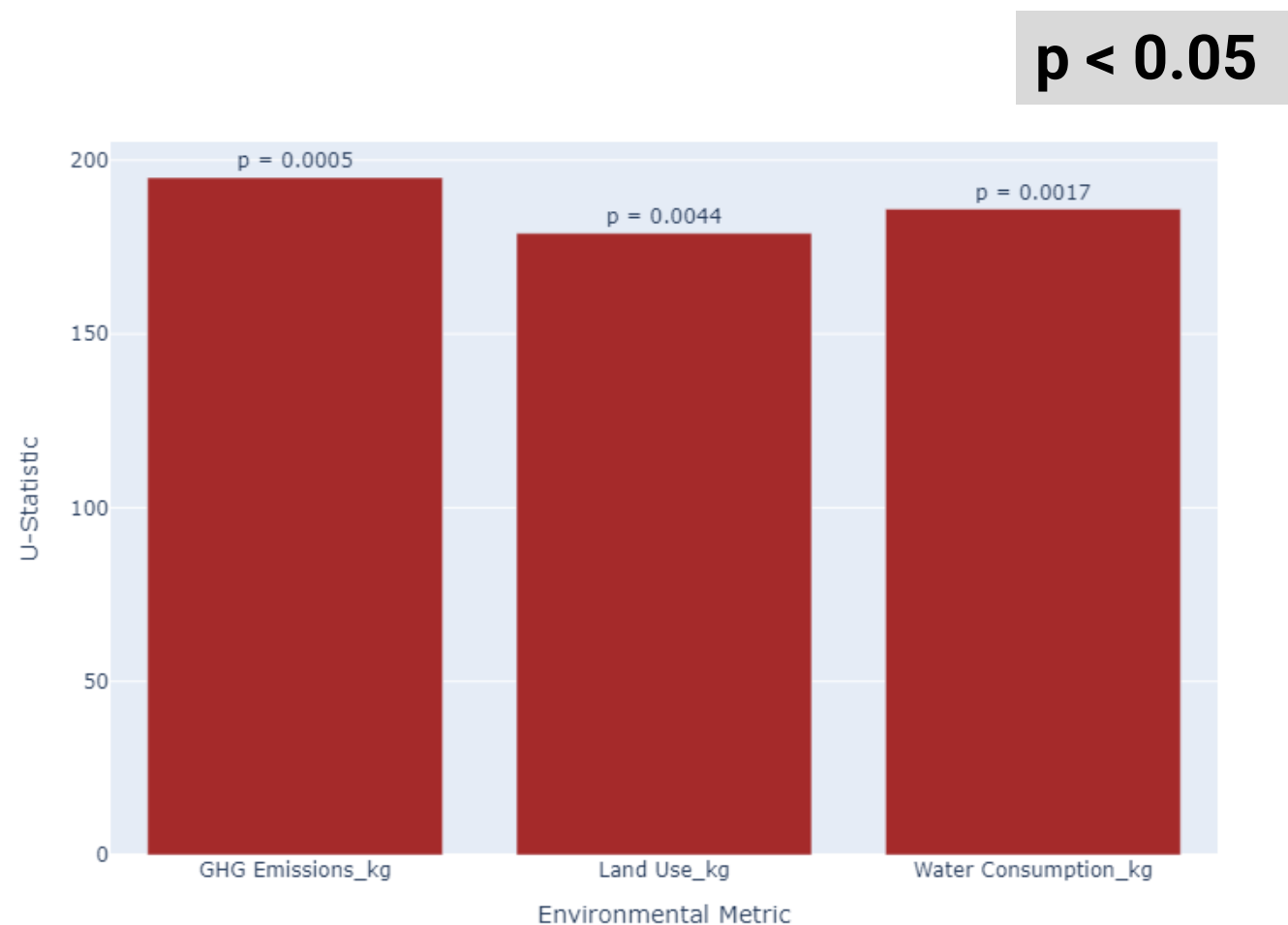
- Animal-based foods consume over 3× more water than plant-based foods.



Comparison of average water use from animal-based and plant-based foods

Are Animal-Based Foods Significantly Worse?

- **Null Hypothesis (H_0)**
 - There is no statistically significant difference in environmental impact between animal-based and plant-based foods.
- **Alternative Hypothesis (H_1)**
 - Animal-based foods have a significantly higher environmental impact than plant-based foods.

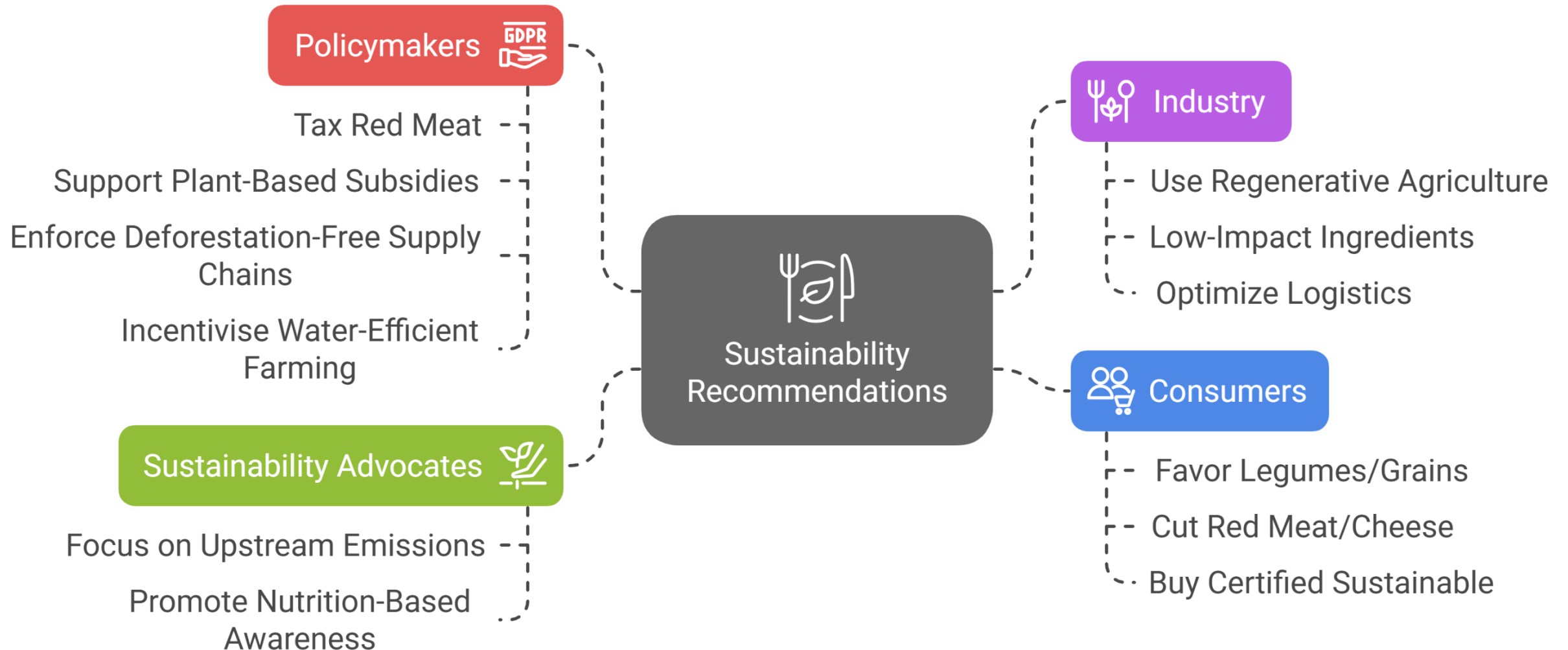


**Mann-Whitney U-Test Results:
Environmental Impact of Animal
vs. Plant-Based Foods**

A collage of various food items including chocolate, tomatoes, apples, lamb chops, raspberries, almonds, yogurt, and salmon. The items are arranged in a circular pattern around the central text.

CONCLUSION AND RECOMMENDATIONS

- **Animal-based foods** are environmentally burdensome, especially beef, lamb, and cheese.
- **Plant-based foods** offer substantial benefits, particularly legumes, grains, and vegetables.
- **Farming, land use change, and feed** are the most impactful supply chain stages.
- **Chocolate, coffee, and nuts** deserve nuanced consideration due to trade-offs.
- Statistical testing confirms **animal-based foods are significantly worse** on all metrics.





- [Link to Medium Article](#)
- [Link to PowerBI Dashboard](#)
- [Link to Full Report Word Document](#)



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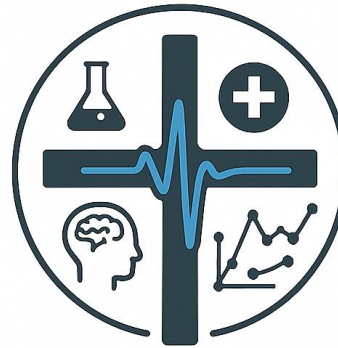
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Project Repo: [Environmental Impact of Food Production](#)

"Data-Driven Solutions for a Sustainable World"



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Thanks for your attention

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